EFFECTIVE REMAINDER EVERYWHERE

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ABSTRACT. We prove that for every modulus $N \geq 1$, every natural number w not divisible by 3, and every natural number r, there is a natural number $n \equiv r \pmod{N}$ whose Collatz orbit reaches w, and that n can be chosen with $n \leq w \cdot 2^{2^{48N^4}}$. In particular, every residue class modulo every positive modulus contains infinitely many predecessors of w. The argument separates the modulus into a 2- and 3-adic part, handled by a parity word of the shortcut map, and a part M coprime to 6, handled by the fibres of the backward closure over $\mathbb{Z}/M\mathbb{Z}$. The fibres are permuted by affine stabilizers, and a minimal-period argument shows that translation by 1 is a stabilizer, so every residue occurs. All results are formalized in Lean 4 using only the core library.

1. INTRODUCTION AND MAIN RESULTS

Let $\mathcal{C}\colon \mathbb{N} \to \mathbb{N}$ be the Collatz map, $\mathcal{C}(n) = n/2$ for even n and $\mathcal{C}(n) = 3n+1$ for odd n. We write $n \rightsquigarrow v$ if $\mathcal{C}^{(k)}(n) = v$ for some $k \geq 0$. In the Lean development, $\mathcal{C}$ is `collatz`, the k-th iterate is `iter`, and $n \rightsquigarrow v$ is `Reaches v n`.

Theorem 1.1 (Effective remainder everywhere)**.** *Let $N \geq 1$, let $w \in \mathbb{N}$ with $3 \nmid w$, and let $r \in \mathbb{N}$. Then there is $n \in \mathbb{N}$ with*

$$n \rightsquigarrow w, \qquad n \equiv r \pmod{N}, \qquad n \leq w \cdot 2^{2^{48N^4}}.$$

Corollary 1.2 (Infinitude in each class)**.** *Under the hypotheses of Theorem 1.1, and for every $L \in \mathbb{N}$, there is $n > L$ with $n \rightsquigarrow w$ and $n \equiv r \pmod{N}$.*

An earlier preprint by the same author, titled *Every Remainder Everywhere* and dated October 29, 2026, showed that witnesses exist in every residue class, but gave no explicit upper bound on the least witness, and handled the factor M by a different argument, based on a finite graph of affine labels and commutators rather than on fibres and stabilizers. The present paper gives the explicit bound of Theorem 1.1 and the argument of Sections 9–11. It supersedes that preprint.

Theorem 1.1 is a specialization of the following bounded form, which carries an explicit lower bound B.

Proposition 1.3 (Bounded form)**.** *Let $N \geq 1$, let $w \in \mathbb{N}$ with $3 \nmid w$, and let $r, B \in \mathbb{N}$. Then there is $n \in \mathbb{N}$ with*

$$B < n, \qquad n \rightsquigarrow w, \qquad n \equiv r \pmod{N}, \qquad n \leq w \cdot 2^{2^{16 \cdot 3N^4 (B+1)}}.$$

Theorem 1.1 says that the least witness is at most $w \cdot 2^{2^{48N^4}}$. The lower bound B in Proposition 1.3 is a device for the proof: setting $B = 0$ gives the exponent $16 \cdot 3N^4 = 48N^4$, and Corollary 1.2 is then derived by changing the modulus rather than by taking B large. Corollary 1.2 follows from Theorem 1.1 by passing to the modulus $N(L+2)$; see Section 13.

The hypothesis $3 \nmid w$ cannot be dropped. If $3 \mid w$, then $3m+1 = w$ has no solution, because $3m+1 \equiv 1 \pmod{3}$. So the only predecessor of a multiple v of 3 is $2v$, which is again a multiple of 3. Hence every $n \rightsquigarrow w$ is divisible by 3. For $N = 3$ and $r = 1$ this rules out every n.

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Every result in this paper is formalized in Lean 4 [2], in a single file `Effective_Remainder_Everywhere_2026-11-01.lean` that imports nothing and uses only the core library. The file was checked with Lean version 4.35.0 (commit `eb6cafb7290e8bd5f2937522d8d804f58f02bafc`); its SHA-256 hash is

`50dd463dbd362fb4b69a9aa37658cc8f119d99954ef685a56be7871ecd2a2db2`.

The only axioms used are `propext`, `Classical.choice` and `Quot.sound`. The file is available at https://github.com/Sio-MLST/Effective-Remainder-Everywhere. Appendix A lists every declaration of the file.

Related work. The parity-vector encoding of Collatz orbits, in which iterates of the shortcut map are written as affine functions of a binary word, was introduced by Terras [5] and appears independently in the work of Everett [3]; it is surveyed by Lagarias [4]. Wirsching [6] studies predecessor sets of the Collatz map and their distribution, including their 3-adic aspects. Applegate and Lagarias [1] use the tree of predecessors to bound densities. The parity words and offsets of Section 4 belong to this encoding. The part of the modulus coprime to 6 is treated in Sections 9–11; the earlier preprint mentioned in the introduction used a different argument for it.

Strategy. Write $N = 2^a 3^b M$ with $\gcd(M, 6) = 1$. The factor $2^a 3^b$ is handled by a parity word, which turns a congruence modulo $2^a 3^b$ into a congruence for a single natural number m that reaches w, modulo $3^{j+b} M$. The factor M is handled by studying the fibres of the closure over $\mathbb{Z}/M\mathbb{Z}$. The fibres are permuted by affine maps, and an argument with minimal periods shows that the translation by 1 is a stabilizer, so every fibre is the whole of $\mathbb{Z}/M\mathbb{Z}$. The statement then follows from an explicit bound on the exponent in a residue-hitting lemma.

2. Conventions

$\mathbb{N} = \{0, 1, 2, \dots\}$, and N denotes a positive natural number. A natural number x is *live* if $3 \nmid x$. Since $3 \mid 0$, the hypothesis $3 \nmid w$ forces $w \geq 1$. For M with $\gcd(M, 6) = 1$, the integers 2 and 3 are invertible modulo M, and $\mathbb{Z}/M\mathbb{Z}$ is identified with $\{0, \dots, M-1\}$ through the reduction map ρ_M.

3. Orbits

Lemma 3.1 (Doubling and transitivity)**.**
(1) For every $a \in \mathbb{N}$, $\mathcal{C}(2a) = a$.
(2) For all m, n, x, $\mathcal{C}^{(m+n)}(x) = \mathcal{C}^{(m)}(\mathcal{C}^{(n)}(x))$.
(3) The relation $\rightsquigarrow$ is transitive.

Lemma 3.2 (Powers of two)**.** *For all w and n, $2^n w \rightsquigarrow w$.*

Proof. Induction on n, using $\mathcal{C}(2 \cdot 2^n w) = 2^n w$ from Lemma 3.1(1). □

Lemma 3.3 (Odd steps)**.** *We have $\mathcal{C}(2y+1) = 6y+4$ and $\mathcal{C}(6y+4) = 3y+2$.*

4. Affine words

The *shortcut map* is $T(n) = n/2$ for even n and $T(n) = (3n+1)/2$ for odd n. A word records a sequence of such steps, the letter 0 standing for an even step and 1 for an odd one.

A *word* is a list $u = [\varepsilon_1, \dots, \varepsilon_\ell]$ with letters in $\{0, 1\}$. The first letter is applied first. Define $|u| = \ell$, and let $|u|_1 = \varepsilon_1 + \cdots + \varepsilon_\ell$ be the number of letters equal to 1. Define the *offset* c_u by $c_{[]} = 0$ and $c_{\varepsilon u'} = 3^{|u'|_1}\varepsilon + 2c_{u'}$.

Lemma 4.1 (Integrality). *Let u be a word and $x \in \mathbb{N}$, and suppose $2^{|u|}$ divides $3^{|u|_1}x + c_u$. Then $x \rightsquigarrow \big(3^{|u|_1}x + c_u\big)/2^{|u|}$.*

Proof. Induction on the word. Since $3^{|u'|_1}$ is odd, the divisibility hypothesis forces $x \equiv \varepsilon \pmod 2$, where ε is the first letter. If $\varepsilon = 0$, then $\mathcal{C}(x) = x/2$. If $\varepsilon = 1$, then x is odd, $\mathcal{C}(x) = 3x + 1$ is even, and $\mathcal{C}\big(3x + 1\big) = (3x + 1)/2$, so x reaches $(3x + 1)/2$ in two steps. In both cases the quotient is the image y of x under T, where $y = x/2$ if $\varepsilon = 0$ and $y = (3x + 1)/2$ if $\varepsilon = 1$, and the induction hypothesis applies to y and the remaining word. □

Lemma 4.2 (Parity words). *For every ℓ and r there is a word p of length ℓ with letters in $\{0, 1\}$ such that 2^ℓ divides $3^{|p|_1}r + c_p$.*

Proof. Induction on ℓ. If $r = 2y$, take the first letter 0 and recurse on y. If $r = 2y + 1$, take the first letter 1 and recurse on $3y + 2 = (3r + 1)/2$. In each case the identity expressing $3^{|p|_1}r + c_p$ as 2 times the corresponding expression for the recursive value carries the divisibility by 2^ℓ up to $2^{\ell+1}$. □

Lemma 4.3 (Size of words). *For every word u with letters in $\{0, 1\}$, $|u|_1 \le |u|$ and $c_u \le 6^{|u|}$.*

5. Arithmetic modulo powers of three

Lemma 5.1. *We have $3 \mid 3^{K+1}$ and $1 < 3^{K+1}$.*

Lemma 5.2 (Live residues). *If $3 \mid m$ and c is live, then $c \bmod m$ is live. A product of live natural numbers is live. Every power of 2 is live, and $4^j \equiv 1 \pmod 3$ for all j.*

Lemma 5.3 (Lifting the exponent for base 4). *For all $K, m \in \mathbb{N}$,*

$$4^m \equiv 1 \pmod{3^{K+1}} \iff 3^K \mid m.$$

Proof. Induction on K. The identity $(1 + 3^j u)^{3^n} = 1 + 3^{j+n}v$ with v live, and the binomial theorem expands $(1 + 3^s v)^n$. Together they show that $4^{3^K} \equiv 1 + 3^{K+1}u$ with u live, which gives the inductive step in both directions. □

Lemma 5.4 (Order of two). *For $K, m \in \mathbb{N}$,*

$$2^m \equiv 1 \pmod{3^{K+1}} \iff 2 \cdot 3^K \mid m.$$

Lemma 5.5 (Injectivity of powers of two). *If $E_1, E_2 < 2 \cdot 3^K$ and $2^{E_1} \equiv 2^{E_2} \pmod{3^{K+1}}$, then $E_1 = E_2$.*

Lemma 5.6 (Two generates the live residues). *If c and d are live and $d < 3^{K+1}$, there is $j \in \mathbb{N}$ with $2^j c \equiv d \pmod{3^{K+1}}$.*

Proof. Suppose not. The map $E \mapsto 2^E c \bmod 3^{K+1}$ takes live values and is injective on $\{0, \dots, 2 \cdot 3^K - 1\}$ by Lemma 5.5. There are exactly $2 \cdot 3^K$ live residues, so the map would be surjective, a contradiction. □

Lemma 5.7 (Cancellation). *If $\gcd(m, a) = 1$ and $ax \equiv ay \pmod m$, then $x \equiv y \pmod m$; the same holds for right multiplication. If c is live, then $\gcd(3, c) = 1$.*

Lemma 5.8 (Parity of exponents). *If $2^n \equiv 1 \pmod 3$ then n is even. Also 3^n is odd.*

6. Residues, invertibility and cardinality

For $M \geq 1$, $\rho_M(n)$ is the class of n in $\mathbb{Z}/M\mathbb{Z}$. The map ρ_M respects sums and products, sends $0, 1, 2, 3$ to the expected elements, and identifies canonical representatives.

Definition 6.1 (Two and three invertible)**.** For $M \geq 1$, write $\mathrm{TI}(M)$ for the statement that 2 and 3 are invertible in $\mathbb{Z}/M\mathbb{Z}$. An element of $\mathbb{Z}/M\mathbb{Z}$ is invertible if it has a multiplicative inverse, and u^{-1} denotes the inverse of an invertible u.

Lemma 6.2. *If* $\gcd(M, 6) = 1$*, then* 2 *and* 3 *are invertible modulo* M*.*

Lemma 6.3. *Invertible factors can be cancelled. If* $\gcd(M, 6) = 1$*, then every power of* 2 *is invertible modulo* M*.*

Definition 6.4 (Cardinality)**.** For a predicate S on $\mathbb{Z}/M\mathbb{Z}$, $|S|$ is the number of elements satisfying it.

Lemma 6.5 (Cardinality estimates)**.** *If* g *is injective and maps* S *into* T*, then* $|S| \leq |T|$*. If* $S \subseteq T$ *and* $|T| \leq |S|$*, then* $T \subseteq S$*, so* $S = T$*. A set of positive cardinality is non-empty, and a non-empty set has positive cardinality.*

7. Congruence tools and orders

Lemma 7.1 (Chinese remainder theorem)**.** *Let* $m, n \geq 1$ *be coprime. For all* $a, b \in \mathbb{N}$ *there is* $y \in \mathbb{N}$ *with* $y \equiv a \pmod{m}$ *and* $y \equiv b \pmod{n}$*. Congruences modulo* m *and modulo* n *combine to a congruence modulo* mn*.*

Lemma 7.2 (Power-of-two period)**.** *If* $M \geq 1$ *and* $\gcd(M, 2) = 1$*, there is* $D \in [1, M]$ *with* $2^D \equiv 1 \pmod{M}$*.*

Proof. The $M + 1$ residues $2^i \bmod M$, for $0 \leq i \leq M$, lie in $\{0, \ldots, M - 1\}$ and cannot all be distinct. Two equal residues with exponents $i < i'$ give $2^{i'-i} \equiv 1$ after cancelling the invertible power 2^i. □

Lemma 7.3 (Factorizations)**.** *Every* $D > 0$ *can be written* $D = 3^\nu d$ *with* $3 \nmid d$*. Every* $N \geq 1$ *can be written* $N = 2^a 3^b M$ *with* $\gcd(M, 6) = 1$*.*

Definition 7.4 (Four-periods)**.** For $m \in \mathbb{N}$, $\mathrm{IsFourPeriod}(M, m)$ means $4^m \equiv 1 \pmod{M}$.

Lemma 7.5. *The four-periods are closed under addition and multiplication, under reduction modulo a four-period, and under gcd. Moreover* $\mathrm{IsFourPeriod}(M, m)$ *holds if and only if* $4^m \equiv 1 \pmod{M}$ *as residues.*

Lemma 7.6 (Four-periods and powers of three)**.** *If* 3^n *is a four-period, there is* $\beta \leq n$ *such that* m *is a four-period if and only if* $3^\beta \mid m$*.*

Proof. Among the four-periods that are powers of 3 and at most 3^n, a minimal one exists by well-ordering. The gcd closure of Lemma 7.5 shows that every four-period is a multiple of it. □

8. The closure and words of bounded length

Definition 8.1 (Closure)**.** Let $Q, w \in \mathbb{N}$. The predicate $\mathrm{Closure}_{Q,w}(x)$ is the least predicate such that

- (base) $x \equiv 2^k w \pmod{Q}$ for some k;
- (doubling) if $\mathrm{Closure}_{Q,w}(z)$, then $\mathrm{Closure}_{Q,w}(2z)$;
- (undoing) if $\mathrm{Closure}_{Q,w}(z)$ and $3x + 1 \equiv 2z \pmod{Q}$, then $\mathrm{Closure}_{Q,w}(x)$;
- (congruence) if $\mathrm{Closure}_{Q,w}(z)$ and $x \equiv z \pmod{Q}$, then $\mathrm{Closure}_{Q,w}(x)$.

Lemma 8.2 (Closure words)**.** *If* $\mathrm{Closure}_{Q,w}(x)$*, then there are a word* u *and* k *with* $3^{|u|_1}x + c_u \equiv 2^{|u|+k}w \pmod{Q}$.

Proof. Induction on the derivation. The base case takes the empty word. Doubling prepends the letter 0, and undoing prepends the letter 1. Each step is checked with the identities of Lemma 4.1. □

Definition 8.3 (Words of bounded length)**.** $\mathrm{HasWord}_{Q,w}(n, x)$ asserts that there is a word u of length at most n and k with $3^{|u|_1}x + c_u \equiv 2^{|u|+k}w \pmod{Q}$.

Lemma 8.4. HasWord *is monotone in* n*, depends only on* $x \bmod Q$*, holds for* $x \equiv 2^k w$ *with the empty word, and is preserved by the backward step: if* $2y \equiv 3^\varepsilon x + \varepsilon \pmod{Q}$ *and* $\mathrm{HasWord}_{Q,w}(n, y)$*, then* $\mathrm{HasWord}_{Q,w}(n + 1, x)$.

Lemma 8.5 (Backward inversion)**.** *Suppose* Q *is odd and* $\mathrm{HasWord}_{Q,w}(n + 1, x)$*. Then either* $\mathrm{HasWord}_{Q,w}(0, x)$*, or there are* $\varepsilon \le 1$ *and* y *with* $\mathrm{HasWord}_{Q,w}(n, y)$ *and* $2y \equiv 3^\varepsilon x + \varepsilon \pmod{Q}$.

Proof. Split the word on its first letter. If the word is empty, the first alternative holds. Otherwise the first letter ε determines y as $\frac{Q+1}{2}(3^\varepsilon x + \varepsilon)$, and the remaining word witnesses $\mathrm{HasWord}_{Q,w}(n, y)$. □

Lemma 8.6 (Stabilization)**.** *There is* $n \le Q$ *such that, for every residue* t*,* $\mathrm{HasWord}_{Q,w}(n+1, t)$ *implies* $\mathrm{HasWord}_{Q,w}(n, t)$.

Proof. Let κ_n be the number of residues t with $\mathrm{HasWord}_{Q,w}(n, t)$. If the conclusion failed, then κ_n would increase by at least one at each step $n \le Q + 1$, which contradicts $\kappa_n \le Q$. □

Lemma 8.7 (Short words)**.** *Suppose* Q *is odd and every residue has some word. Then every residue has a word of length at most* Q.

Proof. Take n as in Lemma 8.6. By Lemma 8.5, $\mathrm{HasWord}_{Q,w}(n+d, x)$ implies $\mathrm{HasWord}_{Q,w}(n, x)$ for every d, and the hypothesis supplies a starting length for each x. □

Proposition 8.8 (Bounded closure words)**.** *Let* $\gcd(M, 6) = 1$*,* $3 \nmid w$ *and* $K \in \mathbb{N}$*. For every* x *there is a word* u *with* $|u| \le 3^{K+1}M$*, and* k*, such that*

$$3^{|u|_1}x + c_u \equiv 2^{|u|+k}w \pmod{3^{K+1}M}.$$

Proof. The modulus $Q = 3^{K+1}M$ is odd. By Proposition 11.9 below, $\mathrm{Closure}_{Q,w}(x)$ holds for every x. Lemma 8.2 gives a word, and Lemma 8.7 shortens it to length at most Q. □

9. Fibres

Fix $M \ge 1$ and $K, w \in \mathbb{N}$, and assume $\mathrm{TI}(M)$ (Definition 6.1) in this section and in Sections 10 and 11; by Lemma 6.2 it holds when $\gcd(M, 6) = 1$. The *fibre* over a is the set of residues modulo M of closure points lying in the class a modulo 3^{K+1}.

Definition 9.1 (Fibre)**.** $\mathrm{Fibre}(M, K, w, a, t)$ holds if there is x with $\mathrm{Closure}_{3^{K+1}M,w}(x)$, $x \equiv a \pmod{3^{K+1}}$ and $\rho_M(x) = t$. With M, K and w fixed we write $\mathrm{Fibre}(a, t)$, and $\mathrm{Fibre}(a)$ for the set of t for which it holds.

Lemma 9.2. *A fibre depends only on* $a \bmod 3^{K+1}$*. If* $t \in \mathrm{Fibre}(a)$ *then* $2t \in \mathrm{Fibre}(2a)$*. Multiplication by* 2 *is injective on* $\mathbb{Z}/M\mathbb{Z}$.

Lemma 9.3 (Equal cardinalities)**.** *For live* a, b*, the fibres over* a *and* b *have equal cardinality.*

Proof. One has $|\mathrm{Fibre}(a)| \le |\mathrm{Fibre}(2^j a)|$ by Lemma 9.2 and injectivity of doubling. Lemma 5.6 gives j with $2^j a \equiv b$, and the argument applies in both directions. □

Lemma 9.4. *For live a,* Fibre$(2a, t)$ *holds if and only if $t = 2s$ for some s with* Fibre(a, s).

Lemma 9.5 (Undo correspondence)**.** *Let $g \equiv 2 \pmod 3$ and let a' be live with $3a' + 1 \equiv 2g \pmod{3^{K+1}}$. Then* Fibre$(a', s)$ *holds if and only if there is t with* Fibre(g, t) *and $3s + 1 = 2t$.*

Proof. For the forward implication, the map $t \mapsto 3^{-1}(2t-1)$ is injective, and Lemma 9.3 compares the two cardinalities. For the reverse implication, the Chinese remainder theorem lifts the witness to the required residues modulo 3^{K+1} and M. □

Lemma 9.6. Fibre$(2^m, t)$ *holds if and only if $t = 2^m s$ for some s with* Fibre$(1, s)$. *For any a with $2^m \equiv a \pmod{3^{K+1}}$ the same holds for* Fibre(a).

10. Stabilizers

Definition 10.1 (Stabilizer)**.** A map $h\colon \mathbb{Z}/M\mathbb{Z} \to \mathbb{Z}/M\mathbb{Z}$ is a stabilizer if Fibre$(1, t) \iff$ Fibre$(1, h(t))$ for every t.

Lemma 10.2. *Stabilizers are closed under composition and inversion, and a map that agrees with a stabilizer everywhere is a stabilizer.*

Lemma 10.3. *If $\gcd(M, 6) = 1$ and $2^m \equiv 1 \pmod{3^{K+1}}$, then $t \mapsto 2^m t$ is a stabilizer.*

Proof. By Lemma 9.6 with $a = 1$, and since 2^m is invertible, Fibre$(1, 2^m t)$ holds if and only if Fibre$(1, t)$ does. □

Lemma 10.4 (Loop stabilizer)**.** *Let $g \equiv 2 \pmod 3$ and let a' be live with $3a' + 1 \equiv 2g \pmod{3^{K+1}}$. Let $2^i \equiv g$ and $2^m \equiv a'$ modulo 3^{K+1}. Then*

$$t \longmapsto 2^{-m} \cdot 3^{-1} \cdot \left(2 \cdot 2^i\, t - 1\right)$$

is a stabilizer.

Proof. Combine Lemma 9.5 for g and a' with Lemma 9.6 for the powers 2^i and 2^m. □

Lemma 10.5. *The map $t \mapsto 3^{-1}(4t - 1)$ is a stabilizer.*

Proof. Apply Lemma 10.4 with $g = 2$, $a' = 1$, $i = 1$ and $m = 0$. The hypotheses hold because $3 \cdot 1 + 1 = 2 \cdot 2$. □

Lemma 10.6 (Affine translations)**.** *Let λ be invertible, and suppose $t \mapsto \lambda t + \mu$ is a stabilizer. Then $t \mapsto t + (\lambda + \mu - 1)$ is a stabilizer.*

Proof. Conjugate the map of Lemma 10.5 by the given stabilizer. The commutator is translation by $\tau = (1 - 4 \cdot 3^{-1})(\lambda + \mu - 1)$. Since $1 - 4 \cdot 3^{-1} = -3^{-1}$, we have $\tau = -3^{-1}(\lambda + \mu - 1)$, so applying the inverse of this translation three times gives translation by $-3\tau = \lambda + \mu - 1$. □

11. Translation periods and full fibres

Definition 11.1 (Translation period)**.** $d \in \mathbb{N}$ is a translation period if $t \mapsto t + \rho_M(d)$ is a stabilizer.

Lemma 11.2. *If $\rho_M(d) = 0$ then d is a translation period. Translation periods are closed under addition, under multiplication by natural numbers, and under subtraction when the result is non-negative.*

Lemma 11.3. *Let $q \mid M$. If q divides the representative of $\rho_M(X) - \rho_M(Y)$, then $X \equiv Y \pmod q$. If q divides the representative of $z \in \mathbb{Z}/M\mathbb{Z}$, then q divides the representative of cz for every $c \in \mathbb{Z}/M\mathbb{Z}$.*

Proposition 11.4 (No non-trivial period)**.** *Let* $q > 1$ *with* $\gcd(q, 6) = 1$*, and suppose* $\mathrm{IsFourPeriod}(q, 3^K)$*. Suppose that for all* g, a', i, m *satisfying the hypotheses of Lemma 10.4 one has* $2^{i+1} \equiv 1 + 3 \cdot 2^m \pmod q$*. Then no such* q *exists.*

Proof. Let $\beta \geq 1$ be the exponent with $\mathrm{IsFourPeriod}(q, m) \iff 3^\beta \mid m$, given by Lemma 7.6. Here $\beta \geq 1$ because q does not divide $4^1 - 1 = 3$. Since 3^{K+1} is odd, $h = (3^{K+1}+1)/2$ is a natural number. Put $g = 2 + 3^{\beta+1}h$ and $a' = 1 + 3^\beta$. These satisfy the hypotheses of Lemma 10.4, and by Lemma 5.6 (with $c = 1$) there are i, m with $2^i \equiv g$ and $2^m \equiv a'$ modulo 3^{K+1}. The hypothesis gives $2^{i+1} \equiv 1 + 3 \cdot 2^m \pmod q$. Since $2^{i+1} \equiv 2g \equiv 1$ and $2^m \equiv a' \equiv 1 \pmod 3$, both $i + 1$ and m are even by Lemma 5.8, so $2^{i+1} = 4^\eta$ and $2^m = 4^\delta$. Modulo $3^{\beta+1}$ one has $4^\eta \equiv 4$, which forces $3^\beta \mid \eta - 1$ by Lemma 5.3, and hence $4^\eta \equiv 4 \pmod q$. Substituting gives $4^\delta \equiv 1 \pmod q$, so $3^\beta \mid \delta$, and therefore $4^\delta \equiv 1 \pmod{3^{\beta+1}}$. But $4^\delta \equiv a' = 1 + 3^\beta \pmod{3^{\beta+1}}$, a contradiction. □

Lemma 11.5 (The period is one)**.** *If* $\gcd(M, 6) = 1$*, then* 1 *is a translation period.*

Proof. The residue of M is 0, so M is a translation period. Let q be the least positive translation period, which exists by well-ordering. Every translation period is a multiple of q, so $q \mid M$. Suppose $q > 1$. The stabilizer $t \mapsto 2^{2 \cdot 3^K} t$ of Lemma 10.3, combined with Lemma 10.6, makes $2^{2 \cdot 3^K} - 1$ a translation period, hence a multiple of q. Thus $4^{3^K} \equiv 1 \pmod q$. The maps of Lemma 10.4 together with Lemma 10.6 supply the remaining hypothesis of Proposition 11.4, which gives a contradiction. Hence $q = 1$. □

Lemma 11.6 (Full fibre over one)**.** *If* $\gcd(M, 6) = 1$ *and* $3 \nmid w$*, then* $\mathrm{Fibre}(1, t)$ *holds for every residue* t*.*

Proof. The fibre over w contains $\rho_M(w)$, and fibres over live residues have equal cardinality, so the fibre over 1 is non-empty. By Lemma 11.5 it is invariant under translation by 1, and a non-empty translation-invariant set of residues is the whole of $\mathbb{Z}/M\mathbb{Z}$. □

Lemma 11.7. *For live* a *and every residue* t*,* $\mathrm{Fibre}(a, t)$ *holds.*

Proof. Lemma 5.6 gives m with $2^m \equiv a$, and Lemma 9.6 transports the full fibre over 1 to the fibre over a. □

Lemma 11.8. *For every* a *and every residue* t*,* $\mathrm{Fibre}(a, t)$ *holds.*

Proof. For non-live a, let $h = (3^{K+1} + 1)/2$, a natural number since 3^{K+1} is odd, and let $g = (3a + 1)h$. Then g is live and $2g \equiv 3a + 1 \pmod{3^{K+1}}$. Fix t. By Lemma 11.7 there is a closure point y with $y \equiv g \pmod{3^{K+1}}$ and $\rho_M(y) = \frac{1}{2}(3t + 1)$, which makes sense because 2 is invertible modulo M. By the Chinese remainder theorem choose x with $x \equiv a \pmod{3^{K+1}}$ and $\rho_M(x) = t$. Then $3x + 1 \equiv 3a + 1 \equiv 2y \pmod{3^{K+1}}$ and $3x + 1 \equiv 2y \pmod M$, so $3x + 1 \equiv 2y \pmod{3^{K+1}M}$. The undoing rule of Definition 8.1 gives $\mathrm{Closure}_{3^{K+1}M,w}(x)$, and so $\mathrm{Fibre}(a, t)$ holds. □

Proposition 11.9 (Every residue is in the closure)**.** *Let* $\gcd(M, 6) = 1$ *and* $3 \nmid w$*. Then* $\mathrm{Closure}_{3^{K+1}M,w}(x)$ *holds for every* x*.*

Proof. By Lemma 11.8 the fibre over $x \bmod 3^{K+1}$ contains $\rho_M(x)$. This gives a closure point x' with $x' \equiv x$ modulo 3^{K+1} and modulo M, hence modulo $3^{K+1}M$. Then the congruence rule applies. □

12. The exponent lemma and residue hitting

Lemma 12.1 (Exponent class)**.** *Let* $\gcd(M, 6) = 1$*,* $3 \nmid w$*, and let* $j, k', Z, D' \in \mathbb{N}$ *with* D' *live,* $2^{3^K D'} \equiv 1 \pmod M$ *and* $Z \equiv 2^{k'} w \pmod{3^{K+1}M}$*. Then for every* B *there is* k *with*

$$B \leq k \leq 2 \cdot 3^{K+j}(k' + 1) + 2 \cdot 3^{K+j}\big(D'^2 \cdot 2 \cdot 3^{K+j}(k' + 1) + 3^K D' B\big)$$

and $Z \equiv 2^k w \pmod{3^{K+j+1}M}$.

Proof. Put $T = 2 \cdot 3^{K+j}$, the order of 2 modulo 3^{K+j+1}. Lemma 5.6 and the periodicity of 2 modulo 3^{K+j+1} give a representative k_2, with $k' \le k_2 < T(k'+1)$, such that $2^{k_2} w \equiv Z \pmod{3^{K+j+1}}$. Since $2^{3^K D'} \equiv 1 \pmod{M}$, the congruence modulo M is preserved by adding multiples of $2 \cdot 3^K D'$. The Chinese remainder theorem then selects $s \in \mathbb{N}$, bounded by $D'^2 T(k'+1)$, for which both congruences hold at once. Taking $k = k_2 + T(s + 3^K D' B)$ gives $k \ge B$ and the stated bound. The explicit constants are checked in `exponent_class`. □

Proposition 12.2 (Hitting a prescribed residue)**.** *Let* $\gcd(M, 6) = 1$, $3 \nmid w$, *and* $b', e, B, X \in \mathbb{N}$ *with*

$$3 \cdot 3^{b'} M^2 \le X, \qquad B \le X.$$

Then there is m *with* $B < m$, $m \equiv e \pmod{3^{b'} M}$, $m \rightsquigarrow w$ *and* $m \le w \cdot 2^{2^{15X}}$.

Proof. Let D be a power-of-two period modulo M (Lemma 7.2), and write $D = 3^\nu d'$ with $3 \nmid d'$ (Lemma 7.3). Set $K = \nu + b'$. Then $2^{3^K d'} \equiv 1 \pmod{M}$, and the size hypotheses give $3^K, d', M \le X$. Apply Proposition 8.8 at depth K to $x = e$. This gives a word u with $|u| \le 3^{K+1} M \le X$ and k' with $3^{|u|_1} e + c_u \equiv 2^{|u|+k'} w \pmod{3^{K+1} M}$. Reducing the exponent modulo $2 \cdot 3^K d'$ and applying Lemma 12.1 with B replaced by $c_u + |u| + 3^{|u|_1}(B+1)$ yields k with $k \le 2^{15X}$, by the size estimate `bound_arith`, and

$$3^{|u|_1} e + c_u \equiv 2^k w \pmod{3^{K+|u|_1+1} M}.$$

Put $m = (2^k w - c_u)/3^{|u|_1}$. This is a natural number, and $3^{|u|_1} m + c_u = 2^k w$. The choice of k gives $m > B$, and the congruence gives $m \equiv e \pmod{3^{b'} M}$. Lemma 4.1 with the word u gives $m \rightsquigarrow 2^{k-|u|} w$, and Lemma 3.2 gives $2^{k-|u|} w \rightsquigarrow w$. Finally $m \le 2^k w \le w \cdot 2^{2^{15X}}$. □

13. Proofs of the main results

Proof of Proposition 1.3. Write $N = 2^a 3^b M$ with $\gcd(M, 6) = 1$ (Lemma 7.3). Let $r' = r \bmod N$, and let p be a parity word of length a for r' (Lemma 4.2), with $j = |p|_1 \le a$ (Lemma 4.3) and $c = c_p$. Then $3^j r' + c = 2^a e$ for some $e \in \mathbb{N}$, and $e \rightsquigarrow r'$ by Lemma 4.1.

Apply Proposition 12.2 with $b' = j + b$, residue e, lower bound $B' = e + 3^j B$, and $X = 3N^4(B+1)$. The size hypotheses hold: $3^j \le N^2$ and $3^b \le N$ give $3 \cdot 3^{b'} M^2 \le 3N^4 \le X$, and $e \le 2^a e = 3^j r' + c \le 2N^3$, because $3^j r' \le N^3$ and $c \le 2^a N^2 \le N^3$. This yields m with $B' < m$, $m \equiv e \pmod{3^{j+b} M}$, $m \rightsquigarrow w$ and $m \le w \cdot 2^{2^{15X}}$.

Write $m - e = 3^j \cdot t$ with $t = 3^b M s$ for some $s \in \mathbb{N}$, and put $n = r' + 2^a t$. Then $3^j n + c = 2^a m$, so Lemma 4.1 applied to p gives $m \rightsquigarrow n$, and hence $n \rightsquigarrow w$ by transitivity. Since $N \mid 2^a t$, we have $n \equiv r' \equiv r \pmod{N}$. Finally $t > B$, because $m > e + 3^j B$, and $n \ge t$, so $B < n$.

For the bound, $n \le 2Nm$, since $r' < N$, $2^a \le N$ and $t \le m$. Since $N < 2^N \le 2^X$ we have $N + 1 \le 2^{15X}$, so $2N \le 2^{N+1} \le 2^{2^{15X}}$. Hence

$$n \le 2Nm \le 2^{2^{15X}} \cdot w \cdot 2^{2^{15X}} = w \cdot 2^{2^{15X+1}} \le w \cdot 2^{2^{16X}} = w \cdot 2^{2^{16 \cdot 3N^4(B+1)}}.$$ □

Proof of Theorem 1.1. Apply Proposition 1.3 with $B = 0$. The exponent $16 \cdot 3N^4(0+1)$ equals $48N^4$. □

Proof of Corollary 1.2. Let $r' = (r \bmod N) + N(L+1)$ and $N' = N(L+2)$. Since $r' < N'$, Theorem 1.1 applied to N' and r' gives $n \rightsquigarrow w$ with $n \bmod N' = r'$. Then $n \ge r' \ge N(L+1) > L$. Since $N \mid N'$, we get $n \equiv r' \equiv r \pmod{N}$. □

Appendix A. Lean names

The table lists, for each section, the Lean names used there and what they denote.

Declarations. Every declaration of the Lean file appears once, including the four constructors of `Closure`.

Section	Lean names	Content
§1, §3	`collatz`, `iter`, `Reaches`, `collatz_double`, `iter_add`, `reaches_trans`, `reaches_pow2`, `collatz_odd`, `collatz_six_mul_add_four`	$\mathcal{C}$, iterates, $n \rightsquigarrow v$, orbit lemmas, odd steps
§4	`oneCount`, `offset`, `integrality`, `integrality_reaches`, `parity_word`, `oneCount_le`, `offset_le`	affine words, integrality, parity words, size bounds
§5	`three_dvd_pow`, `one_lt_pow3`, `live_mod`, `live_mul`, `two_pow_mod_three_ne`, `four_pow_mod_three`, `four_pow_mod_eq_one_iff`, `pow3_form`, `pow_one_add`, `two_pow_mod_eq_one_iff`, `two_pow_inj`, `two_generates_live`, `mod_cancel_mul_left`, `mod_cancel_mul_right`, `coprime3`, `pow2_even_of_mod3`, `three_pow_odd`	arithmetic modulo powers of three
§6	`residue`, `residue_val`, `residue_eq_iff`, `residue_add`, `residue_mul`, `residue_zero`, `residue_one`, `residue_two`, `residue_three`, `residue_self_val`, `residue_self`, `residue_val_self`, `TwoThreeInvertible`, `unitInv`, `mul_unitInv`, `two_mul_unitInv`, `three_mul_unitInv`, `unitInv_eq_of_mul`, `IsInvertible`, `isInvertible_mul`, `isInvertible_unitInv`, `twoThreeInvertible_of_coprime6`, `exists_inv_residue`, `exists_inv_of_coprime`, `cancel_invertible`, `twoPow_invertible`, `cardinality`, `mem_cardlist`, `cardinality_le_of_inj`, `cardinality_eq_of_sub`, `cardinality_pos`, `exists_of_cardinality_pos`	residues, invertibility, cardinality
§7	`crt_exists`, `mod_combine`, `mod_combine_le`, `mod_of_dvd_mod`, `mod_eq_of_le_sub`, `mod_add_left_cancel`, `mod_lin`, `mod_mul_right_cong`, `pow_mod_period`, `exists_pow2_period`, `split_three`, `split_two_three`, `coprime6_of_not_dvd`, `IsFourPeriod`, `isFourPeriod_add`, `isFourPeriod_mul`, `isFourPeriod_cancel`, `isFourPeriod_mod`, `isFourPeriod_gcd`, `isFourPeriod_iff`, `eq_three_pow_of_dvd`, `exists_min_pow3`, `exists_order_pow3`, `exists_least`, `two_mul_inj`, `three_pow_le`, `six_pow_le`	congruence tools, periods, factorizations, orders
§8	`Closure`, `Closure.base`, `Closure.dbl`, `Closure.undo`, `Closure.cong`, `closure_word`, `HasWord`, `hasWord_mono`, `hasWord_le`, `hasWord_mod`, `hasWord_base`, `hasWord_step`, `hasWord_inv`, `cardinality_le_modulus`, `hasWord_stable`, `hasWord_all_short`, `closure_word_short`, `SameMod`	closure, words of bounded length, stabilization
§9	`Fibre`, `fibre_mod`, `fibre_dbl`, `fibre_cardinality_dbl_le`, `fibre_cardinality_pow_le`, `fibre_cardinality_le_live`, `fibre_cardinality_const`, `fibre_dbl_iff`, `fibre_undo_iff`, `fibre_pow`, `fibre_live`	fibres and their cardinalities

Section	Lean names	Content
§10	`Stabilizer`, `stabilizer_comp`, `stabilizer_inv`, `stabilizer_ext`, `stabilizer_mul`, `stabilizer_loop`, `stabilizer_special_loop`, `stabilizer_affine_translation`	stabilizers
§11	`IsTranslationPeriod`, `translationPeriod_of_eq`, `translationPeriod_add`, `translationPeriod_mul`, `translationPeriod_sub`, `translationPeriod_one`, `mod_of_dvd_sub`, `dvd_val_mul`, `period_contra`, `fibre_one_full`, `fibre_full_live`, `fibre_full`, `closure_all`	translation periods, minimal period, full fibres
§12	`exponent_class`, `hit_residue`, `bound_arith`, `pow2bound_mul`, `pow2bound_le`, `le_two_pow_self`, `pow2_period_lift`	exponent lemma, residue hitting, size estimates
§13	`remainder_bounded`, `effective_remainder_everywhere`, `remainder_infinitely_often`	main results

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